كلية الحاسبات و الذكاء الصناعي

جامعة حلوان
Helwan University

Slothy

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- ## Acknowledgments

First and foremost, we would like to extend our special thanks to

Dr. Safaa Magdy who supervised our graduation project and guided us through the different stages of our academic journey. Your guidance, patience, and constructive comments have been invaluable in helping us grow academically and personally. She pushed me to think critically, pushed me to explore innovative solutions, and gave us directions for overcoming obstacles. We are really grateful for your unwavering support and belief in our potential.

Throughout our time at the College, Drs have played a pivotal role in shaping my education and fostering our passion for computer science and artificial intelligence. Your dedication to teaching, your vast knowledge, and your willingness to go the extra mile to ensure our success has had a significant impact on our teaching experience.

- **Project Timetable**

Task Name	***Duration***	***Start***	***Finish***
Initiation	6 days	20/11/22	27/11/22
Plan	2 days	20/11/22	23/11/22
Proposal	3 days	24/11/22	27/11/22
Documentation	11 day	28/11/22	8/3/22
Phase 1	6 day	28/11/22	4/12/22
Phase 2	16 day	5/12/22	21/12/22
Phase 3	21 day	22/12/22	1/1/23
Phase 4	8 day	4/2/23	12/2/23
Phase 5	9 days	13/2/23	22/2/23
Phase 6	22 day	23/2/23	1/3/23
Phase 7	2 days	2/3/23	4/3/23
Phase 8	4 days	5/3/23	9/3/23
Implementation	9 days	9/3/23	18/5/23
Front-end	19 days	9/3/23	28/3/23
Back-end	23 day	1/4/23	24/4/23
Database design	19 days	25/3/23	15/4/23
Integration	2 days	16/5/23	18/5/23
Delivering	11 day	19/5/23	1/6/23
Testing	6 days	19/5/23	25/5/23
Fix Bugs	5 days	26/5/23	1/6/23
Deliver the project	0 day	13/6/23	13/6/23

- **UX/UI Design Process**

The ui/ux design process includes several steps, each step representing a distinct stage in the iterative design cycle.

1.1 Research.
1.2 Define the user problem.
1.3 Design.
1.4 Prototype.
1.5 Validate.
1.6 Build.
1.7 QA Test.
1.8 Launch.

1.1 Research

- At this point, we have done a lot of surveys and questionnaires which helped us a lot in the design phase, which also helped us extract rich data and ideas.

- This also helps tailor your app roadmap based on the needs of your users.

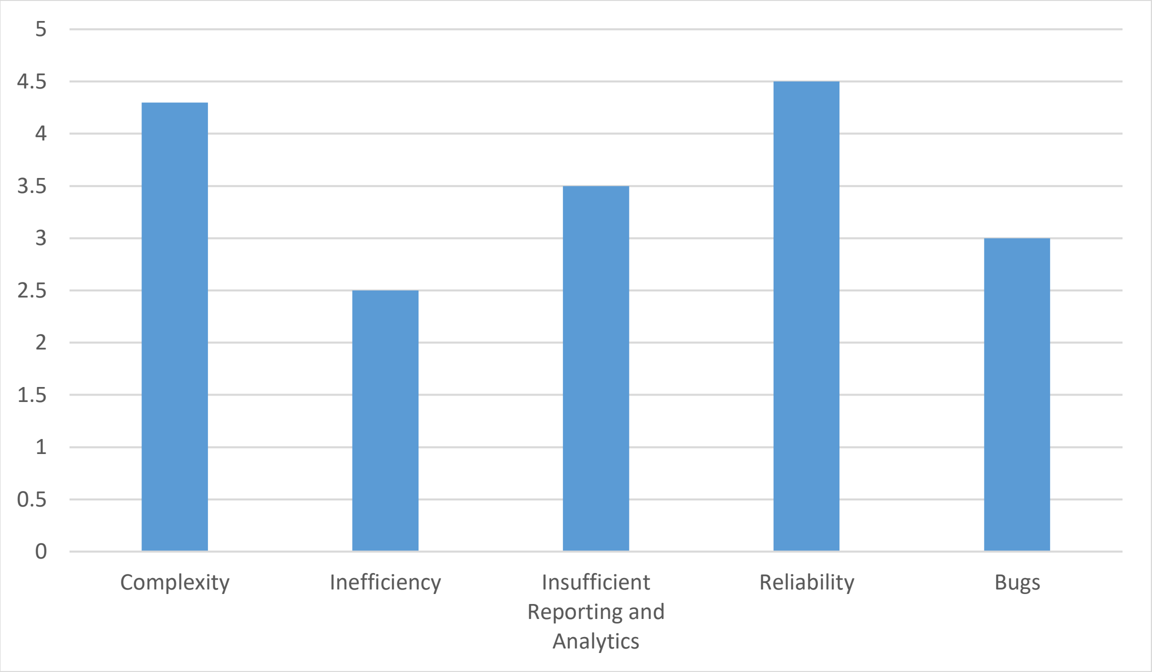
- Catch negative feedback before bad ratings come in.

- Some of the questions that were asked to the users:

• How do you currently manage your time and schedule?
• What are your biggest challenges or frustrations when it comes to managing your time effectively?
• What specific features or functionalities would you like to see in a time management application?
• Do you prefer a visual representation of your schedule (e.g., calendar view, timeline view)?
• How do you prioritize tasks and activities in your daily or weekly schedule?
• How do you prefer to receive reminders and notifications?

1.2 Define the user problem.

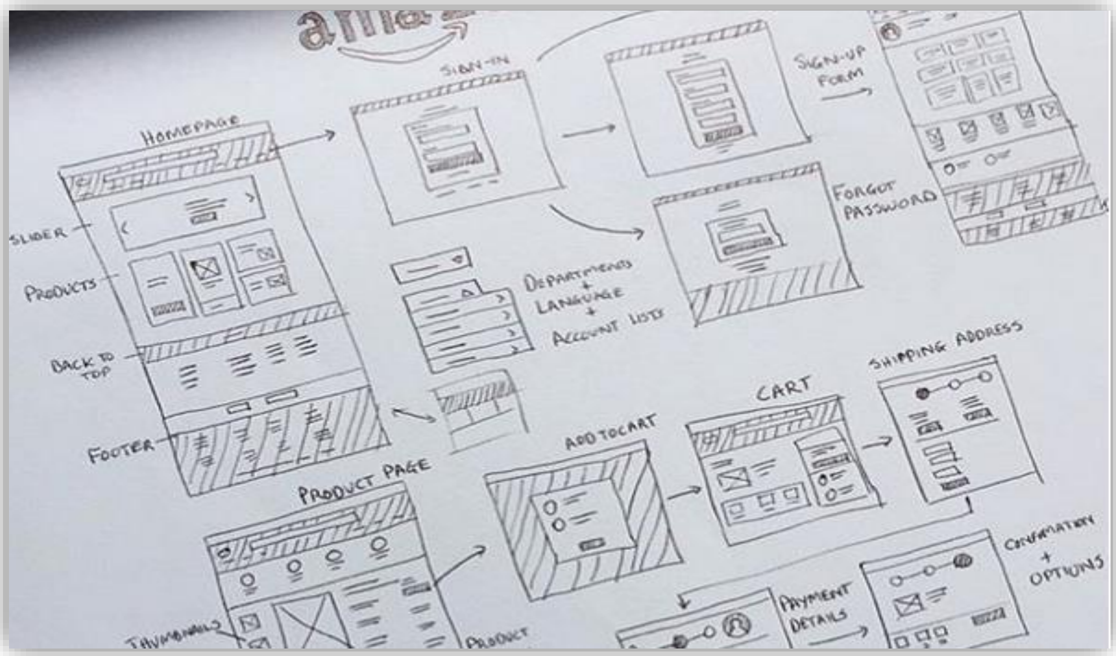
- In this step, the user problem or challenge is clearly defined based on the insights and findings from the research phase. It involves compiling data and identifying specific issues or opportunities that need to be addressed through design.



1.3 Design

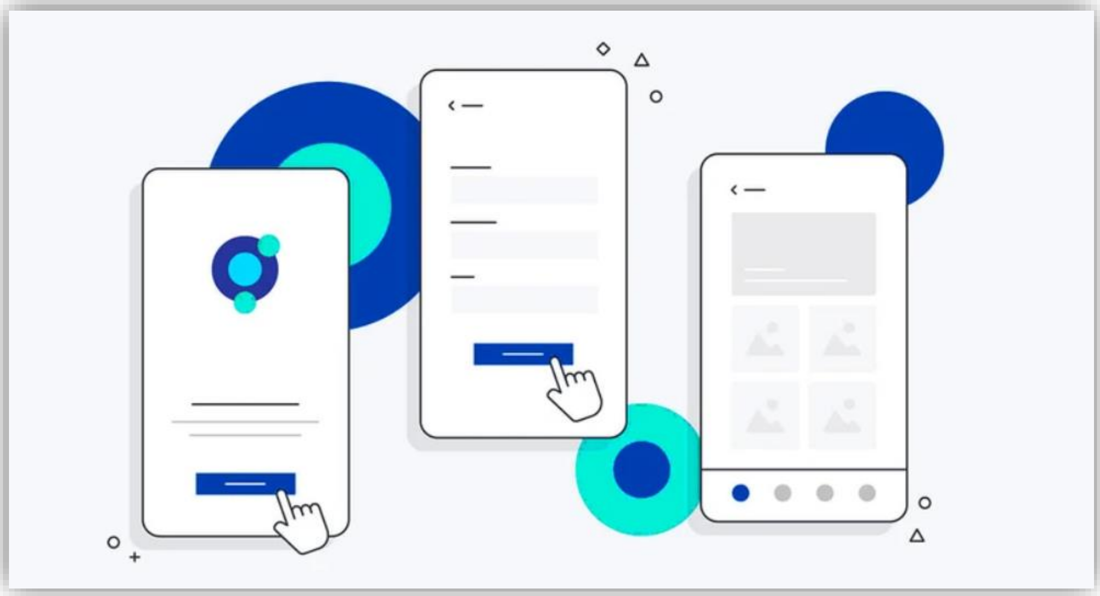
- The design stage begins with the drawing. We have drawn hand drawings to visualize the concept in simple terms and to facilitate the design process.

- We designed the individual components, such as buttons, icons, forms, etc., ensuring consistency and ease of use, where the focus is on different colors, fonts, and layouts to catch the user's attention.



1.4 Prototype

- In this step, we have created a functional but simplified version of the design to provide a concrete representation of the proposed solution.



1.5 Validate

- We implemented this step by validating the prototype with real users to gather feedback and ideas.
- We have collected user feedback as usability testing, user interviews and other evaluation methods are used to improvement before the final product goes live, and delivers this feedback from the user's point of view
- The feedback obtained is used to iterate and improve the design.

1.6 Build

- Once the design has been validated and refined, the development or implementation phase begins.

- This step involves translating the design into a fully functional product or system, using programming languages, frameworks, and other technical tools.

1.7 QA Test

- We have performed Quality Assurance (QA) testing to ensure that the application meets the required specifications, works correctly, and is free from defects or issues.

- Includes rigorous testing to fix bugs and improve performance.

1.8 Launch

- The final step involves launching the app for users.

- It involves publishing, distributing and communicating the newly built solution to the target audience.

Abstract

- Managing your time effectively is an important professional skill to develop. Organizing your tasks each day helps you complete work on time.

- Having strong time-management skills can ultimately lead to accomplishing key goals and advancing in your career.

- Is essential for any successful individual as it allows you to balance work and personal life while still achieving your goals.

- can set priorities, break down complex projects into smaller, manageable tasks, and track progress in real-time, They can prioritize tasks, set deadlines, and categorize activities based on their individual requirements.

- Our everyday task app offers features like convenience, customization, productivity, integration, accountability, motivation and room for innovation.

Chapter 1: Introduction

In this chapter we are going to discuss and go deeper in the overview of the project and know more about its scope and limitations and explain some terminologies we will find throughout the document.

Chapter Headlines:

- **1.1** Overview
- **1.2** Objectives
- **1.3** Purpose
- **1.4** Scope
- **1.5** General Constraints

1.1 Overview

- First, our application is an application that helps the user organize their daily tasks according to priority in an effective and orderly manner.

- Second, includes everything you need to reach your goals, wrapped in a beautiful and intuitive interface. It motivates you, helps you stay focused, and keeps you on track. It is for all the little things that make a big difference.

- Third, with deadlines approaching and a busy schedule, it can be hard to remember to practice good habits like meditation, walking or drinking enough water. Our app tracks your habits and monitors progress by setting a goal to challenge yourself.

- No need to wade through all your habits to figure out what to do, Habit List only shows what's due today. and you can set the default order for your habits, so items that you'll complete first are at the top of your list. Tap today and you will see your completion percentage for the day.

- you can create project time blocks for each day, drag and drop blocks throughout your calendar and time how long each project takes you.

- The app can include rewards, badges, leaderboards, or challenges that encourage users to complete tasks consistently and stay productive.

1.2 Objectives

- Use English as the main language in the application to help all different people with many cultures to use the application easily.

- Get motivated with streaks ,keeping streaks alive is a powerful motivator. See how high you can go, then try to beat your personal best.

- Set daily personal tasks to achieve, focus on important goals first, remind you through notifications to stay on track, and manage time easily.

1.3 Purpose

It allows creating a schedule and rooms and classifying them according to the tasks you enter and helping to implement them effectively and well.

1.4 Scope

The time management application, increase productivity and organize daily tasks, allows you to take and upload your notes and access them from anywhere in addition to organizing and taking notes.

1.5 General Constraints

- Indiscipline "Human factor" like being late in delivering tasks or attending meetings.
- Hesitation, especially when it is related to taking a serious step or learning a new technique.
- Time Management.
- Learning new technologies may take much time.

Chapter 2: Project "Planning and analysis"

In this chapter we are going to analyze and study the app also defining its strengths that solve some issues in other apps.

Chapter Headlines:

2.1 Project Planning

2.1.1 Feasibility Study

Feasibility is defined as the practical extent to which a project can be performed successfully. To evaluate feasibility, a feasibility study is performed, which determines whether the solution considered to accomplish the requirements is practical and workable in the software.

*** Operational feasibility**

The application will be easy to use and make it easier for the users to carry out and organize the daily tasks. It will also be safe for users and will maintain their privacy.

*** Technical feasibility**

The hardware and software for developing, maintaining, and launching the application will be reliable and available to us. the application will not be the need for training users. The system will be maintainable in the future and easy to connect with external systems.

*** Economic feasibility**

The software will be developed by ourselves. We don't need any cost of the fees and license also, we don't have any training costs for users.

*** Schedule feasibility**

Of course, we divided the application development stages into small tasks with a specific working duration with a specific deadline.

2.1.2 Estimated Cost

We may need some cost to buy software and datasets to help our project expand its scope. The development of any project must be organized and have a clear time plan for the entire work of the project.

2.2 Analysis and Limitation of existing system

Notion

What is Notion?

Is a powerful all-in-one workspace and note-taking application. It allows you to create and organize various types of content, including text, images, files, to-do lists, calendars, and databases, among others.

Who use notion?

- Enterprise
- Education
- Managers
- Small business
- Personal use
- Remote work

Defects

Notion is a powerful tool that offers numerous benefits to its users, it has some drawbacks you should know before deciding whether to use it.

- Overwhelming to use: There are so many different types of blocks and templates available that it can be hard to know where to start. This can make it difficult to use Notion efficiently, especially if you are new to the platform.

- Limited mobile app: Notion's mobile app is not as robust as its desktop version. Some features may not be available on the mobile app, which can limit its functionality for users who rely heavily on mobile devices.

New Features

- Gallery View: This feature allows users to view their databases in a gallery format, presenting the information as cards with images.

- Dark Mode: Notion added a dark mode option, allowing users to switch to a darker color scheme for reduced eye strain and improved readability in low-light environments.

- Performance Improvements: Notion has focused on optimizing its performance, making the application faster, more responsive, and capable of handling larger databases and complex projects.

Asana

What is Asana?

Asana makes it easy to get your most important work done. Increase efficiency to deliver results and hit your goals on every project.

Asana lets you put all your plans and projects in one place so you can work productively without the chaos and confusion.

Who use notion

- Students
- Company-wide
- IT Department
- Project Management

Defects

- Because Asana lacks time-tracking capabilities, you'll need to use an external time-tracking program.

- With so many duties in one location, the process might get slow.

New Features

- Timeline View: Allows users to create and visualize project schedules and dependencies in a Gantt chart-like interface. Provides a clear overview of project schedules, deadlines, and task dependencies.

- Portfolios and Dashboards API: Asana has provided an API for Portfolios and Dashboards, allowing developers to build integrations and extract data from these features.

24me

What is 24me?

24me is an award-winning Personal Assistant that helps millions of people all over the world to boost their productivity.

It is an easy-to-use and yet super powerful app that puts everything related to your schedule in one place: your calendar, to-do list, notes and Personal Accounts.

24me saves you time for the things that matter most in your life.

Defects

- User Interface: Some users may find the user interface of 24me to be complex.
- Task Management Complexity: While 24me aims to provide an all-in-one solution, some users might find the task management features to be complex or overloaded.

New Features

- Combines task management
- Smart and sophisticated calendar
- Beautiful design that makes it so fun to use

Relationship between the Related Work and our work

Features / App Name	Slothy	Notion	Asana	24me
Dark Mode	✗	✓	✗	✗
Timeline View	✗	✓	✓	✗
Dashboards API	✗	✓	✓	✓
User-friendly UI/UX	✓	✗	✗	✗
Task Creation and Management	✓	✓	✓	✗
Reminders and Notifications	✓	✓	✓	✓
Task Notes	✓	✓	✓	✓
Priority Tasks	✓	✓	✓	✓
Integrate pages	✓	✓	✓	✓
Ease of Use	✓	✗	✓	✗
Smart Design	✓	✓	✓	✓

2.3 Need for the new system

- The old system is not flexible in dealing with the user and the user finds it difficult to use, in addition to some Features that you must make the site strong and reliable.
- Adding the possibility for the user to all these new features that make it easier for him to organize his daily tasks well and add some important notes that make the site reliable in use.

2.4 Analysis of the new system

2.4.1 User requirements

- User can think, write and plan through the application.
- User arranges tasks according to priority.
- User can create and edit pages by creating a simple personal task list.
- User find pages using the search bar and customize it to meet his needs.
- User can write some notes. There is a button called import so that it can import and edit any file example PDF and modify it.
- User can use the Timer to complete the task at a specific time.
- User can use the Reminder to send a notification for completing tasks at a specific time.
- User can divide the tasks into several small tasks.
- User can collect coins, and this helps him to communicate with others who have the same interests.

2.4.2 System requirements

- Flexibility of use and movement between the pages of the site.
- Manage tasks and do it exactly the way you want.
- There is no difficulty in using the site.
- The speed of response to the user to rely on modern systems.
- Save time and effort for the user.
- Helps you fight procrastination, and get things done today.
- It gives you complete control over your tasks.
- It helps in capturing, organizing, searching and finishing tasks and notes in a good way.

2.4.3 Domain Requirements

- Easy to use.
- Does not require training.
- Is understandable for all users and is also comfortable during use.
- The performance of our application is high in terms of the quality of the result.
- The speed of response.
- The speed of recovery from any failure.

2.4.4 Functional Requirements

Use Case Name	Signup
Pre-condition	User Can create a new Account for the first time with (name, password, e-mail)
Description	The Usert's first step in the application which will allow him/her to enter the application for the first time after filling in the data.
Post-condition	The user has successfully entered into the system for the first time.
Use Case Name	Login
Pre-condition	User enters his/her username and password
Description	User can login in system if he has an account after the authentication process has been performed.
Post-condition	The User is logged in successfully and redirected to the home page.

Use Case Name	View Profile
Pre-condition	User is logged in
Description	The user can view his tasks and all personal data on his profile such as completed tasks.
Post-condition	Profile page will open.

Use Case Name	Edit Profile
Pre-condition	User is logged in and the profile page is opened
Description	User are allowed to set and change which tasks they are interested in and which need to be performed.
Post-condition	Return to the profile page with new data.

Use Case Name	View Tasks
Pre-condition	User opens his/her profile
Description	User can view all Tasks that he/she completed.
Post-condition	All the user’s tasks are displayed.

Use Case Name	Add Task
Pre-condition	User opens the home page
Description	User can add a new task
Post-condition	The new task has been added successfully.

Use Case Name	Search
Pre-condition	User opens the home page
Description	User can search for tasks that he has completed, he can also search for tasks that he has not finished yet to be completed.
Post-condition	The user can search for successfully completed and incomplete tasks.

Use Case Name	Set Timer
Pre-condition	User opens the home page
Description	User can set a timer to finish his tasks at a specific time.
Post-condition	Successful completion of tasks in the specific time.

Use Case Name	Set Calendar
Pre-condition	User opens the home page
Description	User can use the calendar to keep track of his task schedule.
Post-condition	Remind the user of all their tasks successfully.

Use Case Name	Set Notes
Pre-condition	User opens the home page
Description	User can write some important notes.
Post-condition	User can refer to these notes at any time.

Use Case Name	Priority Tasks
Pre-condition	None
Description	User arranges tasks according to priority.
Post-condition	It makes it easier for the user to carry out tasks with minimal effort.

Use Case Name	tasks dividing
Pre-condition	User opens the home page
Description	User can divide the tasks into several small tasks.
Post-condition	It makes it easier for the user to carry out tasks with minimal effort.

2.4.5 Non- Functional Requirements

- **Correctness**: The result of the functions should be accurate and pure by validating or testing the system.
- **Reliability**: The application has the ability of fast recovering from failure.
- **Maintainability**: Is the ability to maintain system bugs and issues with less loss that doesn't affect the services or stop it. With ability of detecting these bugs and solving it.
- **Performance**: The performance of our application is high in terms of the quality of the result, the speed of response, and the speed of recovery from any failure.
- **Usability**: Easy to use, does not require training, is understandable for all users and is also comfortable during use.
- **Availability**: Easy to access and available when needed can be used in any time.
- **Portability**: The application can work on many different platforms because it is developed by Flutter.
- **Security**: The application protect user data, system identification, system authentication.
- **Capacity**: The maximum number of the concurrent users is unlimited users.

2.5 Advantages of the new system

- **<u>Do you ever feel overwhelmed or anxious about all you have to do, but things never get done?</u>** Slothy helps you fight procrastination and get daily tasks done well and done on time.
- Make the system easier and more efficient with features.
- User can organize their daily tasks according to priority in an effective and orderly manner.
- The app helps the user in mental training to operate his mind.
- The app divides the big task into several small ones.
- Get timely reminders about upcoming deadlines to help keep you on track.

2.6 Risk and Risk Managements

- **Misdiagnosis of malfunctions**: we must put some possibilities, not just one possibility.
- **Risk of Quality**: You must make sure of reliable sources
- Not communicating with team members permanently we resort to working in parallel, Conducting online meetings to discuss problems that are expected to occur periodically.
- **Software problem**: The lack of sufficient educational resources and a lot of research to find sufficient resources to start the project.

- **Late delivery of the tasks** : A deadline is set for each task so that the project is delivered at the required time.

Chapter 3: Software Design

In this chapter we are going to discuss and go deeper in application's design and present its diagrams and database.

Chapter Headlines:

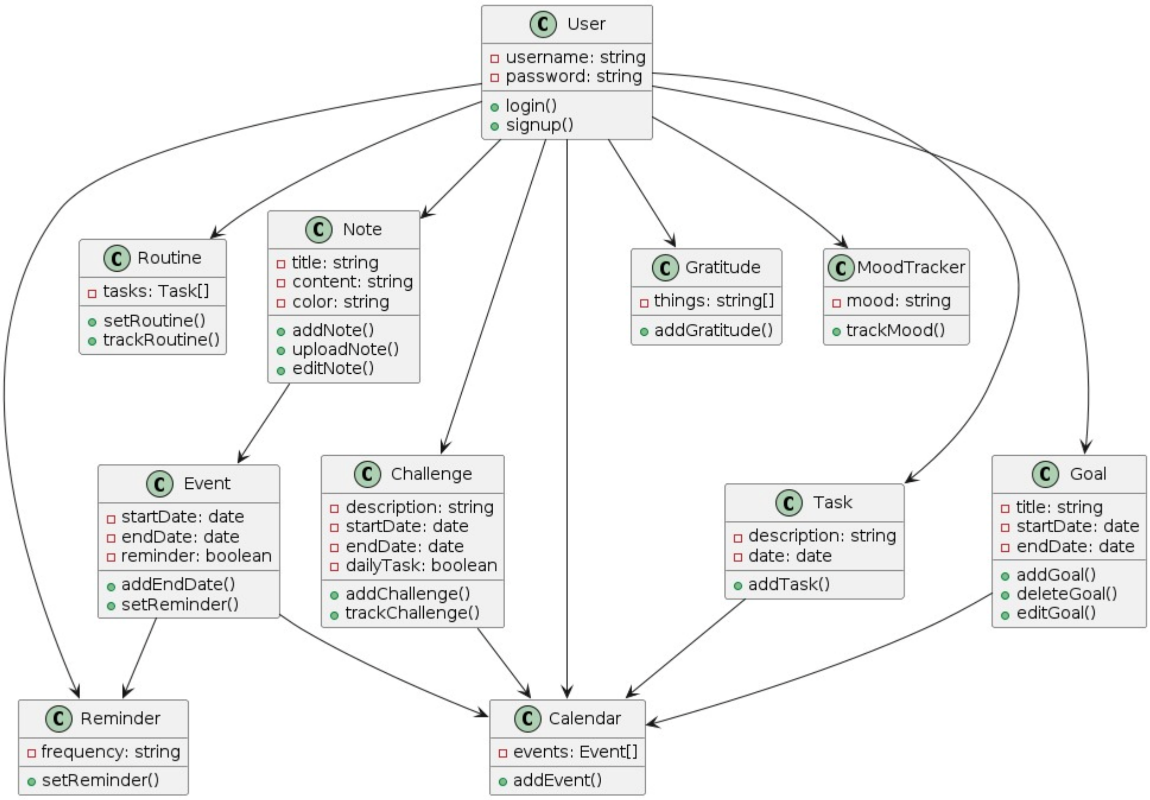
3.1 Design of database (ERD or Class) Diagram

3.2 Use case diagram

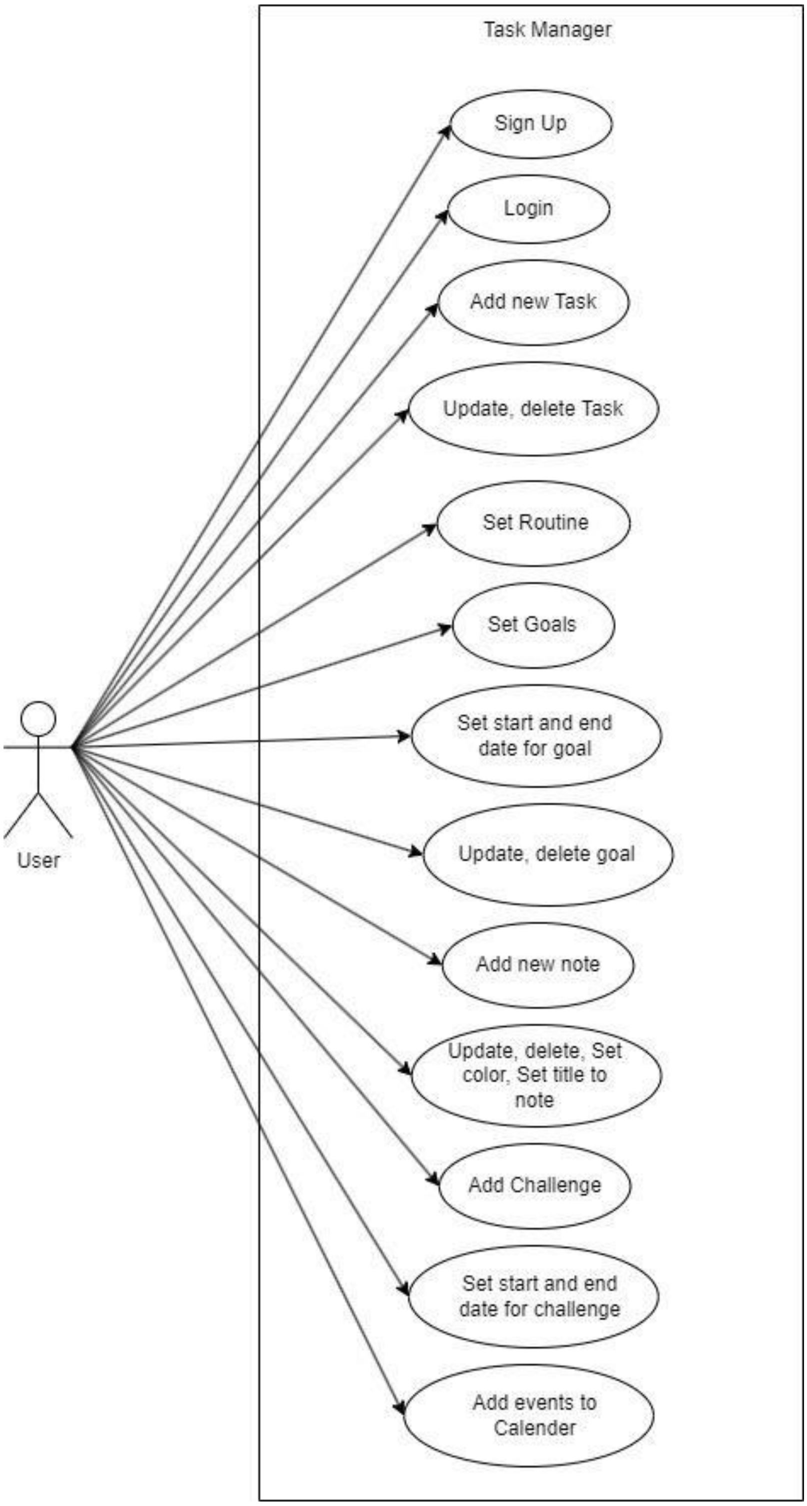
3.4 sequence diagram

3.5 activity diagram

3.1 Design of database Class Diagram

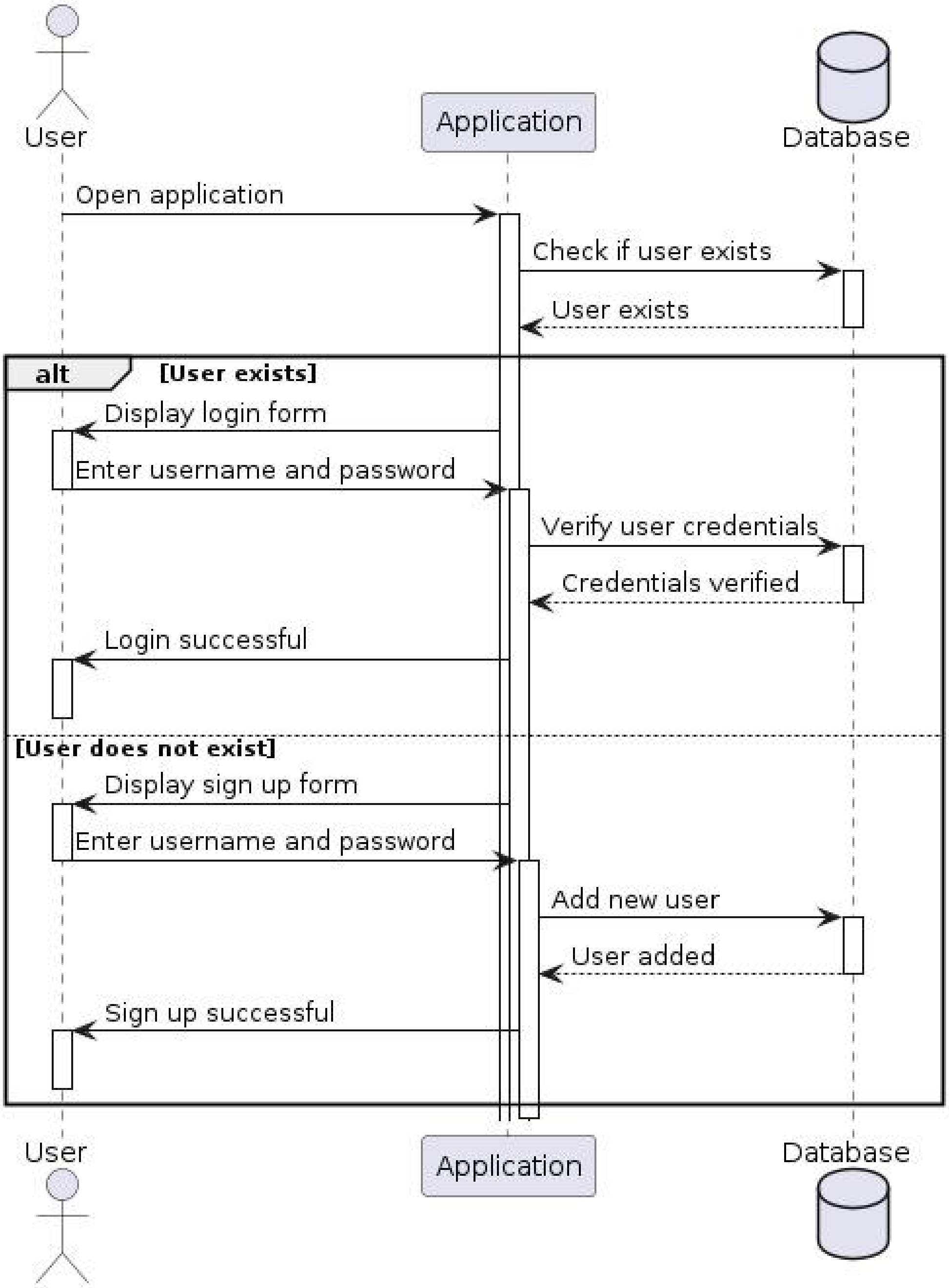


3.2 Use case diagram

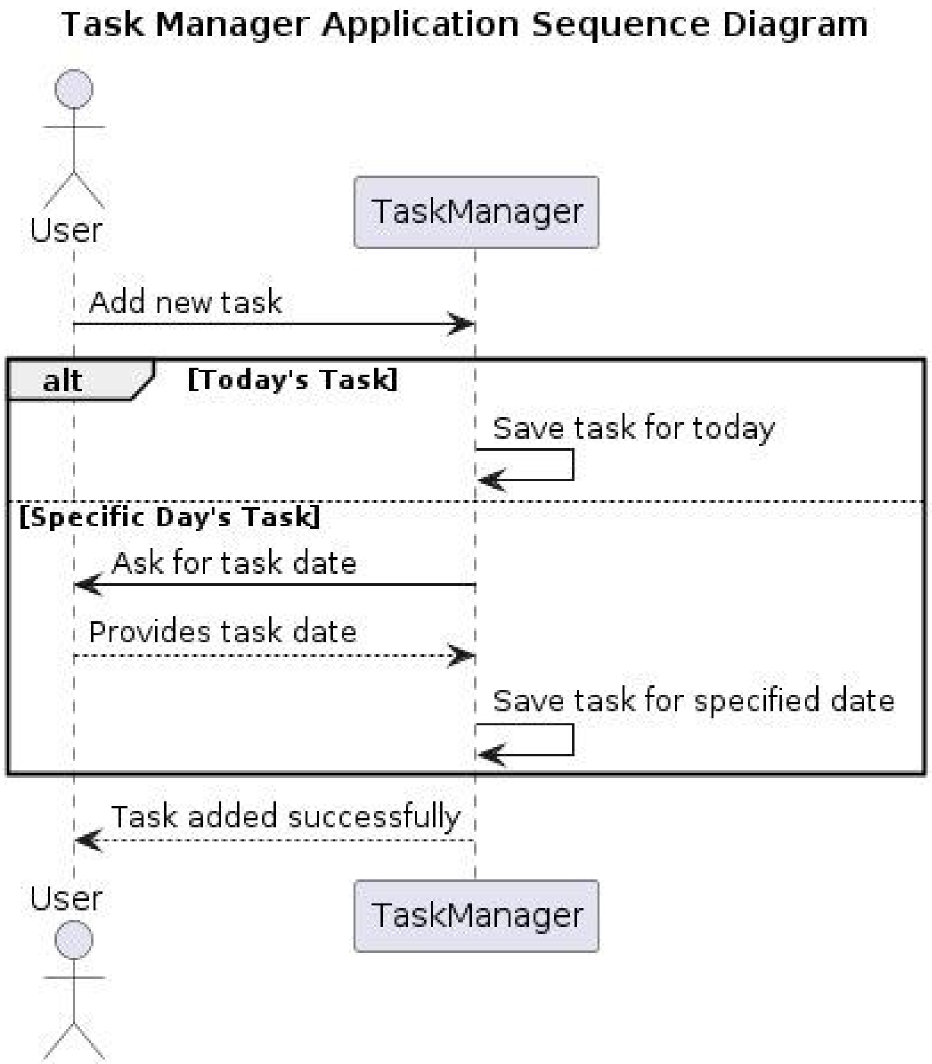


3.4 sequence diagram

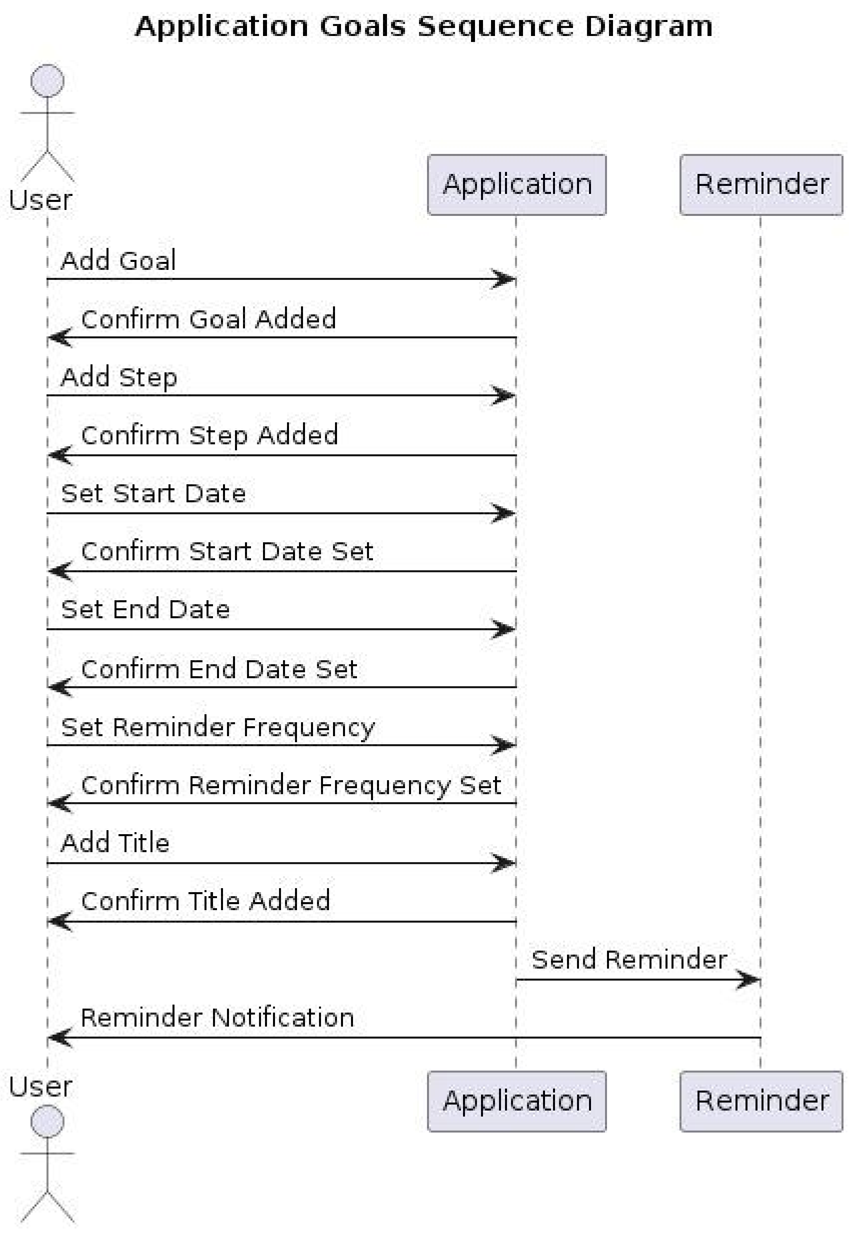
-Login/ sign up sequence diagram



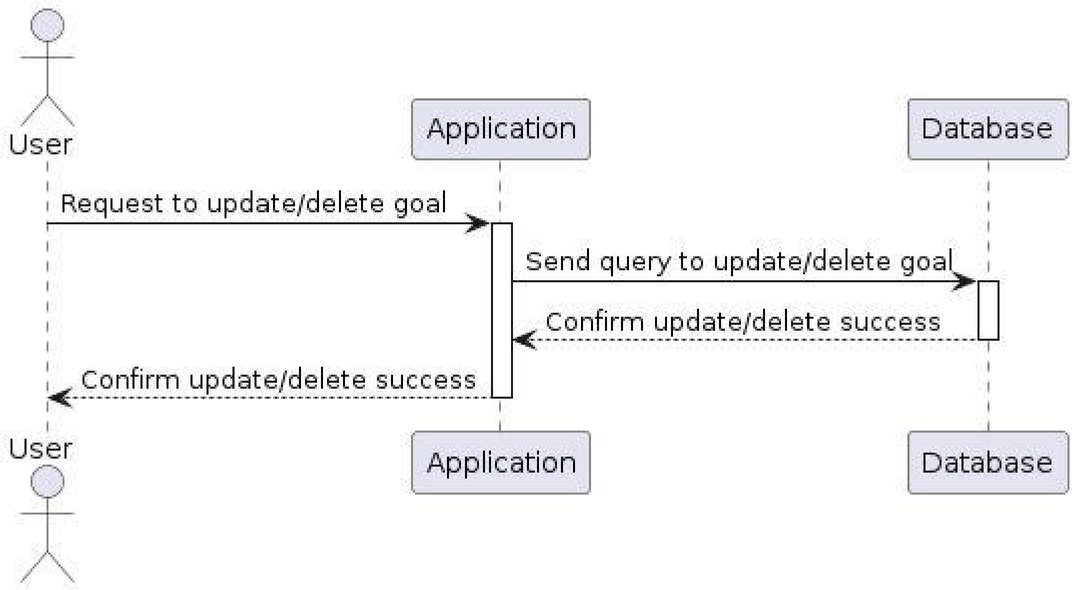
-Add task sequence diagram



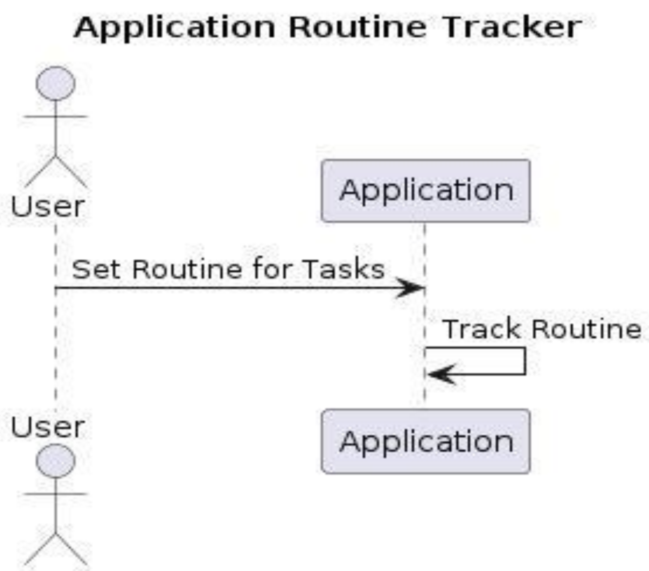
-Add goals sequence diagram



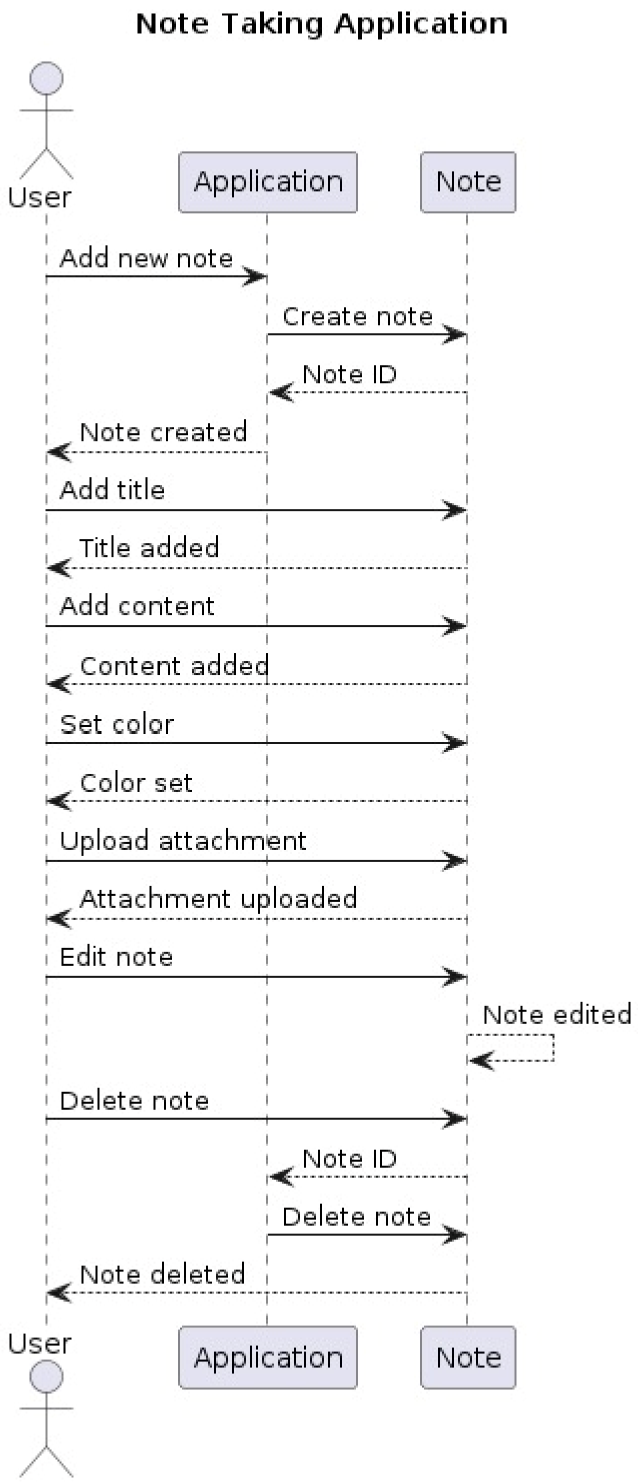
-Update / delete goals sequence diagram



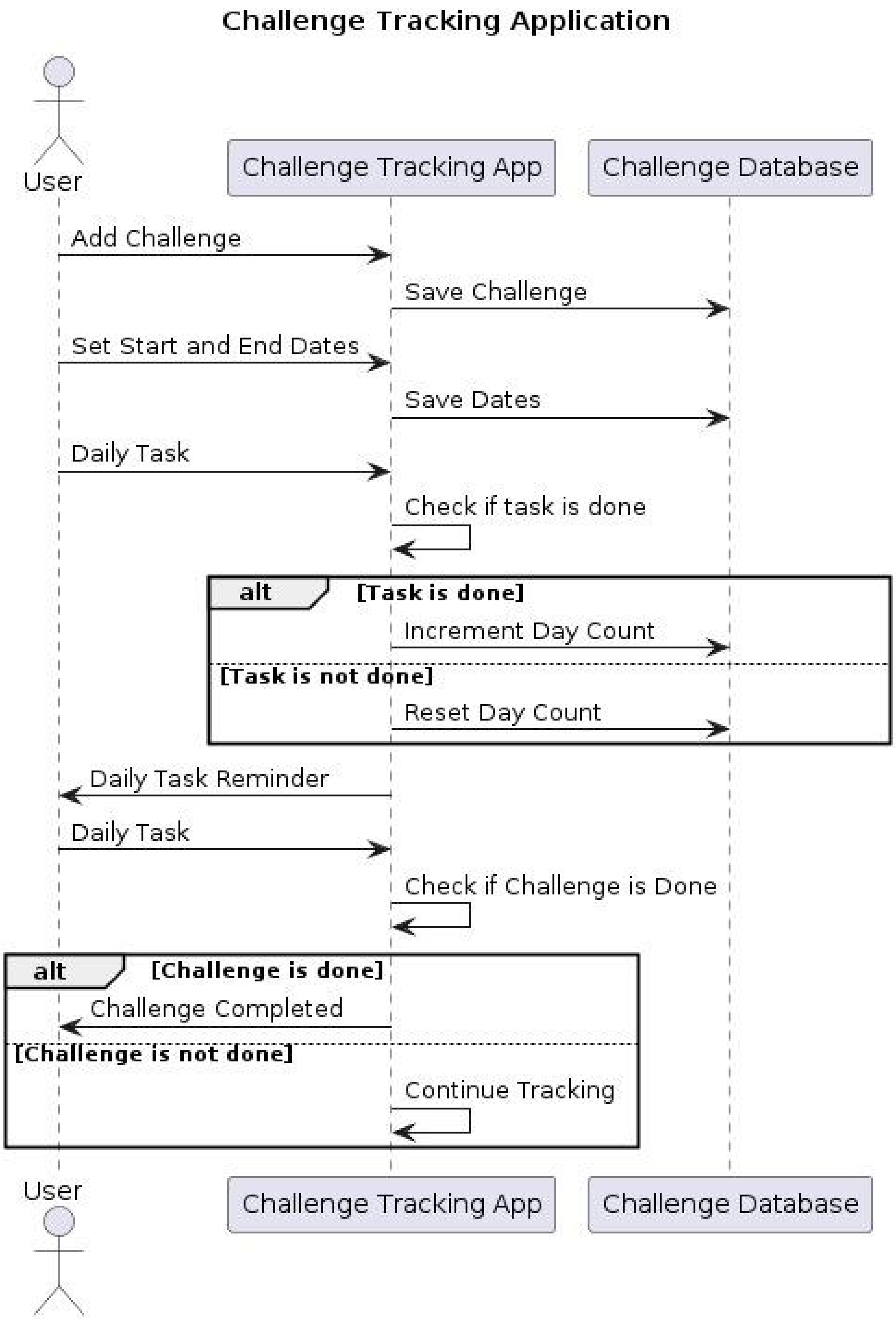
-Set routine sequence diagram



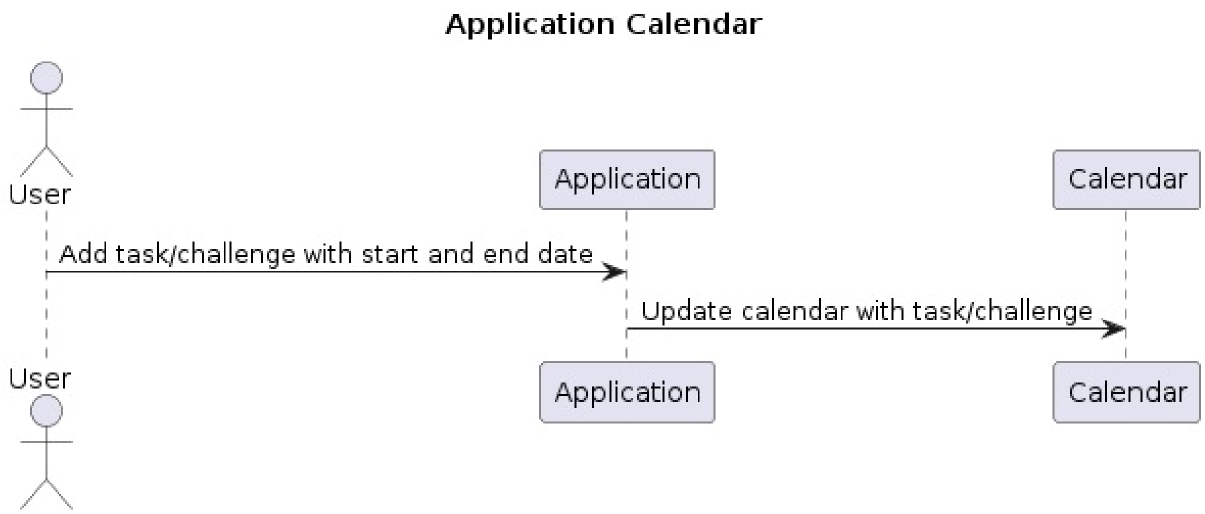
-Add note sequence diagram



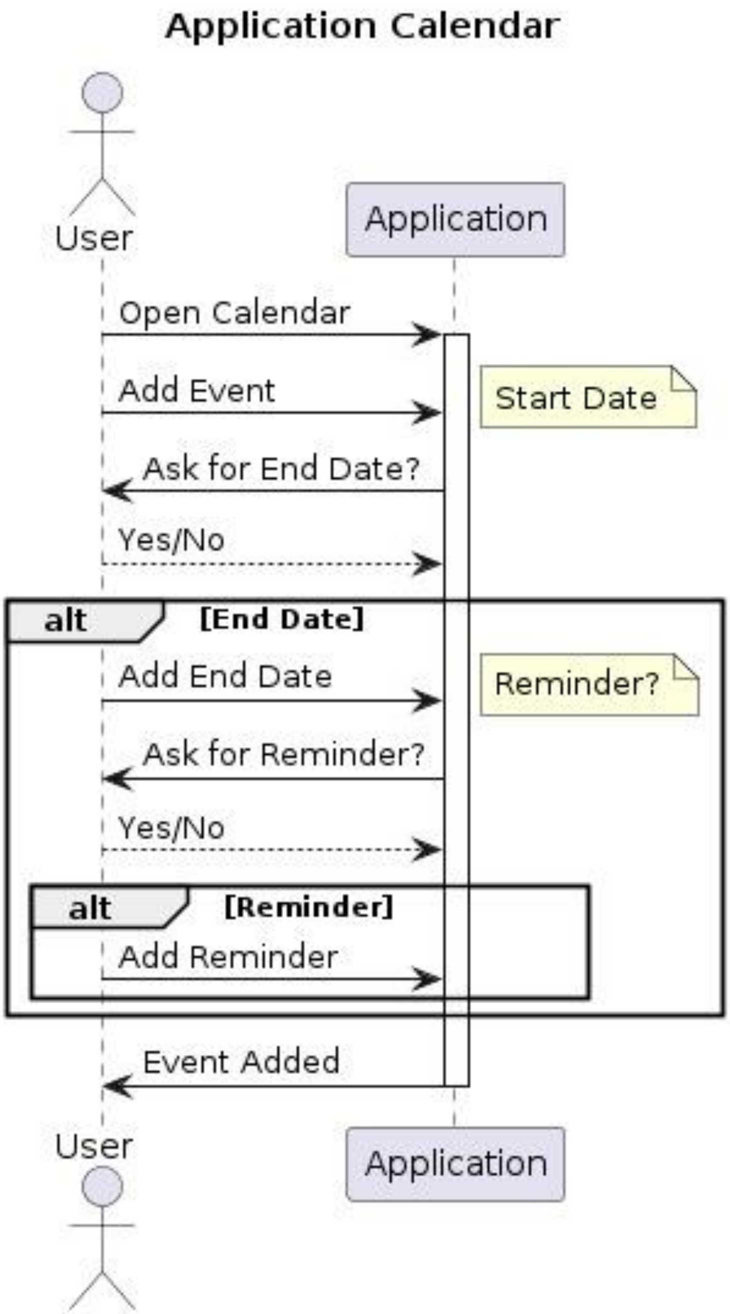
-Challenge tracking sequence diagram



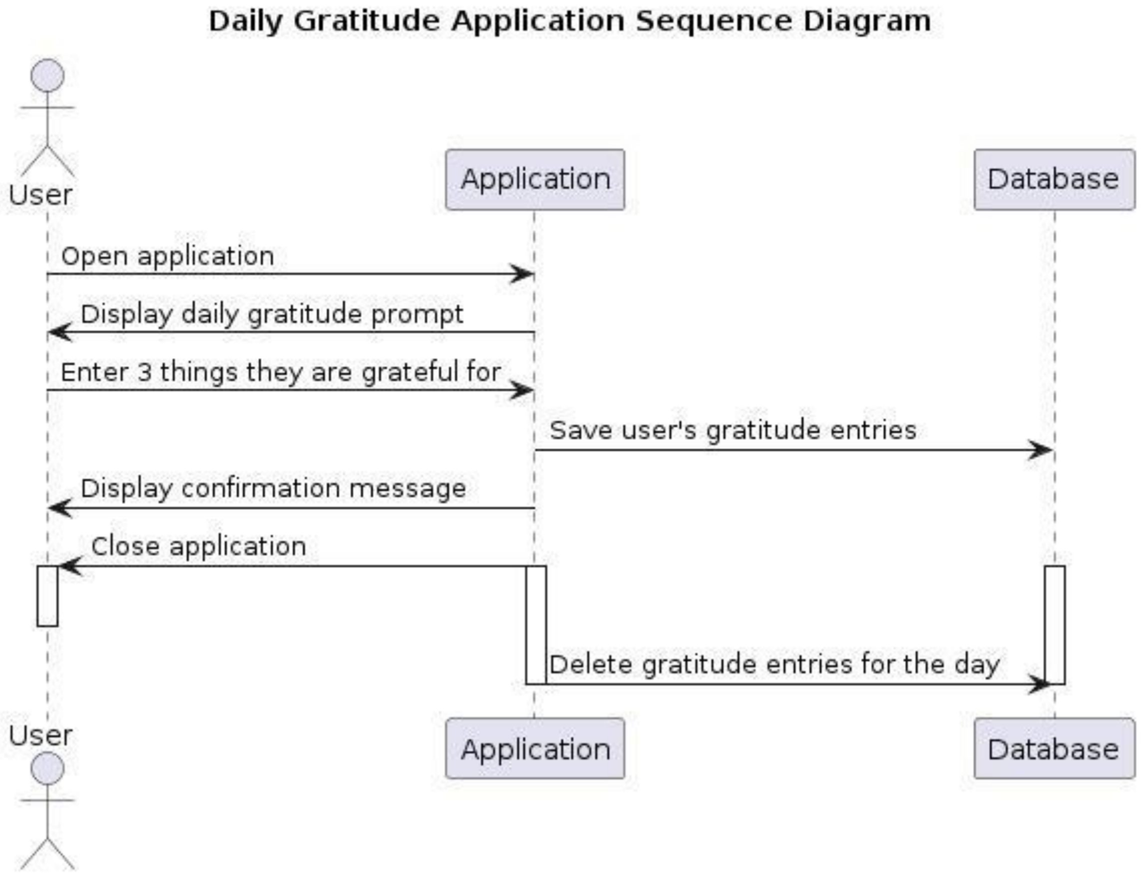
-Update calendar sequence diagram



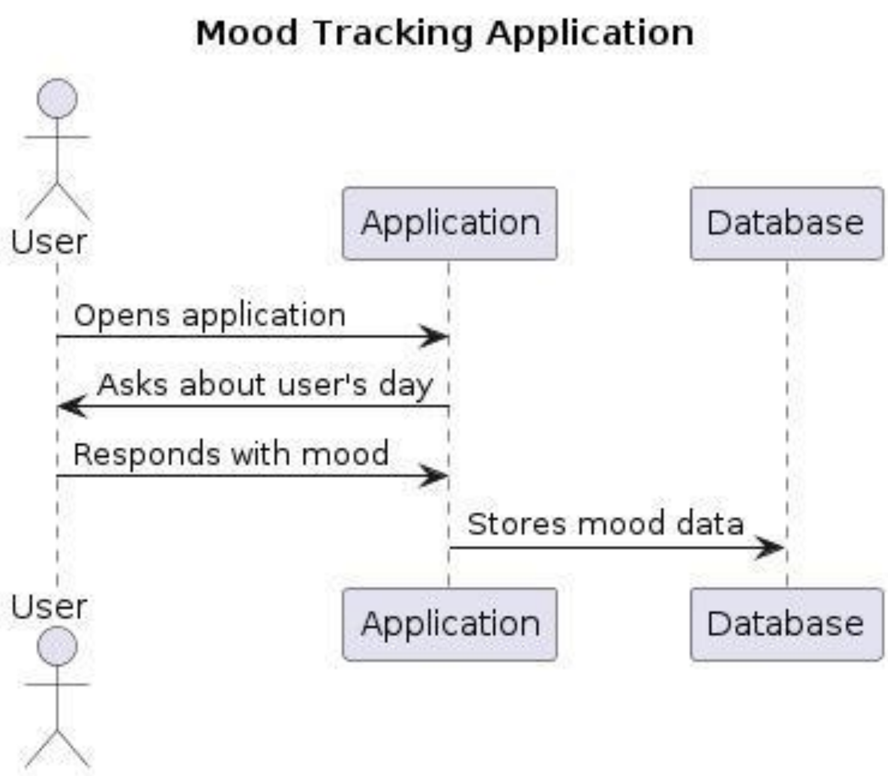
-Calendar sequence diagram



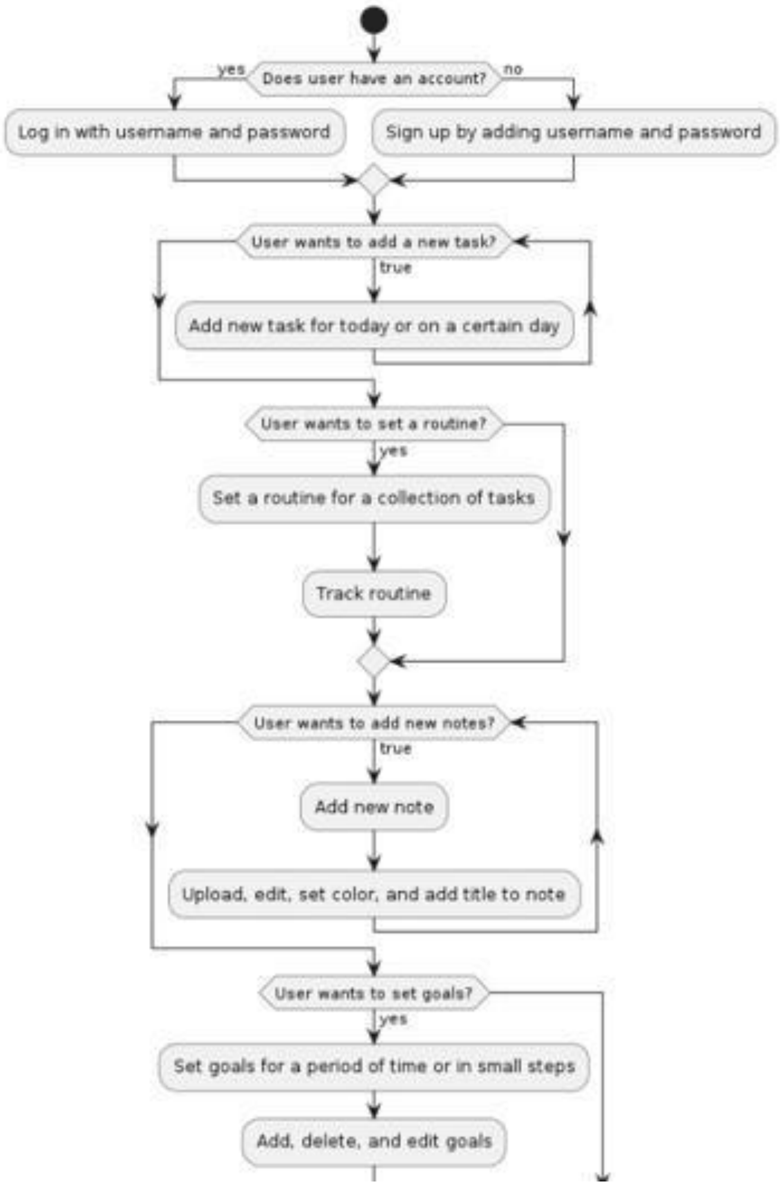
-Gratitude sequence diagram



-Mood tracking sequence diagram



3.5 activity diagram




Set start and end date for goals and steps
Remind user of goals daily, weekly, or monthly
User wants to list 3 things they are grateful for?
true
Ask user about 3 things they are grateful for
User wants to track their mood?
true
Ask user about their mood
User wants to add challenges?
yes
Add challenges
Track daily task for challenge
Set start and end date for challenge
Remind user to do daily task

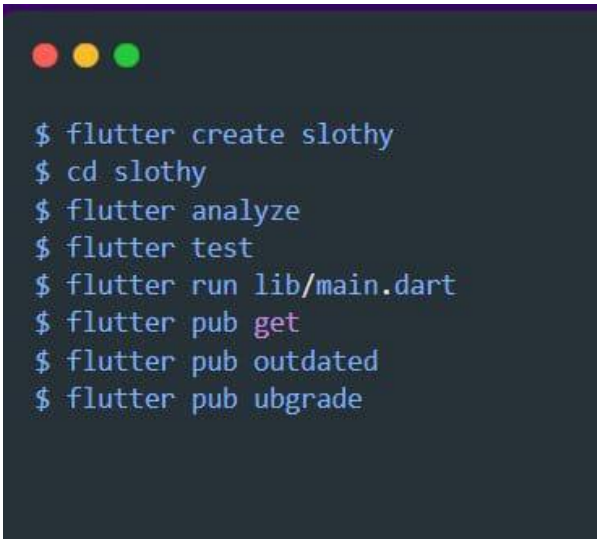
Chapter 4: Implementation

In this chapter, we will discuss the process of developing an implementation plan. It includes implementing the steps, strategies, or procedures necessary to achieve the desired result or goal. Implementation typically follows planning and preparation phases, during which the necessary resources, schedules, and responsibilities are identified.

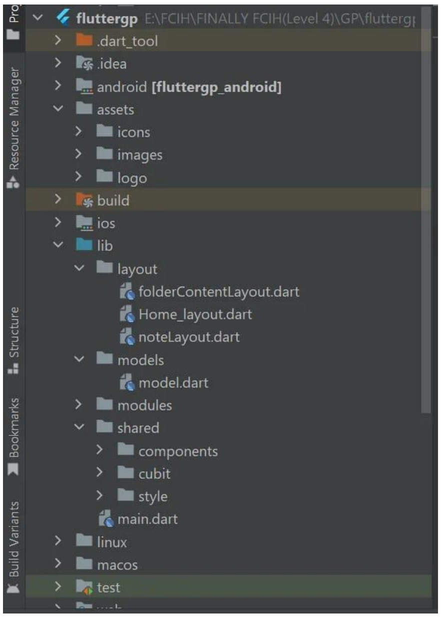
Mobile Implementation:

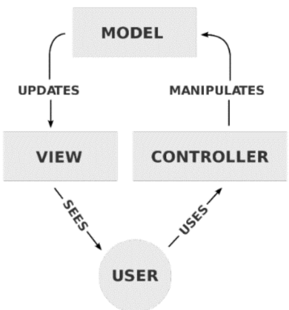
- The technology and the framework we used to develop Slothy mobile application is Flutter.

1. Flutter Commend



2. After Creating our project, we need to choose App Architecture and we use MVC design Pattern The Model View Controller (MVC) design pattern specifies that an application consist of a data model, presentation information, and control information.





3. We need to define the third-party packages that we will need in our application

```yaml
cupertino_icons: ^1.0.2
dropdown_button2: ^2.1.0
firebase_core:
firebase_messaging:
firebase_auth:
google_sign_in:
flutter_facebook_auth:
http:
google_fonts:
animated_splash_screen: ^1.3.0
curved_navigation_bar: ^1.0.3
bottom_navy_bar: ^6.0.0
persistent_bottom_nav_bar: any
google_nav_bar: ^5.0.6
table_calendar: ^3.0.9
intl: ^0.18.1
anim_search_bar: ^2.0.3
```

4. Then We need to initialize Firebase in main method.

```dart
Future<void> main() async{
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp();
  FirebaseMessaging.onBackgroundMessage(_firebaseMassagingBackgroundHandeler as BackgroundMessageHandler);

  await
  FirebaseMessaging.instance.setForegroundNotificationPresentationOptions(
    alert: true,
    badge: true,
    sound: true,
  );
  runApp(const MyApp());
}
```

5. Then We need to create our initial Route and initial Theme

```
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return BlocProvider(
      create: (BuildContext context) => AppCubit(),
      child: BlocConsumer<AppCubit, AppStates>(
        listener: (context, state) {},
        builder: (context, state) {
          return MaterialApp(
              debugShowCheckedModeBanner: false,
              theme: ThemeData(),
              darkTheme: ThemeData(),
              themeMode: AppCubit.get(context).isDark
                  ? ThemeMode.dark
                  : ThemeMode.light,
              home: AnimatedSplashScreen(
                duration: 3000,
                splash: splashScreen(),
                nextScreen: welcomeScreen(),
                splashTransition: SplashTransition.rotationTransition,
                backgroundColor: Colors.cyan,
              )
          );
        },
      ),
    );
  }
}
```

Back-end Implementation

- As a start point we have section of User who Access Nodes in Application ,and we will take it with some Functions in code for back end and firebase Database.

- We use "add function" to add goal and get attributes such as (title , way and reminder ...) from user to add new records in firebase database.

```
void addNewGoal(
  String uid,
  String goalTitle,
  String description,
  String rule,
  String way,
  String startDate,
  String endDate,
  String reminde,
  ) async {
    // A goal entry.
    final postData = {
        'title':goalTitle
        'description': goalTitle
        'rule':rule
        'way':way
        'startDate': startDate
        'endDate': EndDate
        'remind': remind
        'crumbTitle': crumbTitle
      }

};
```

- This Function make sure for user that who get data from firebasedatabase such from goal node after add new goal and condition if user get records of goals data a massage say " a data record exist " else " data record not exist"

```
final ref = FirebaseDatabase.instance.ref();
final snapshot = await ref.child('goals/$goalId').get();
if (snapshot.exists) {
    print(snapshot.value);
} else {
    print('No data available.');
}
```

- we use "add function" to add crumbs to allow user to achieve The main goal after that user make sure that all crumbs finished or not and get attributes such as (title , way and startDate ..endDate ...) from user.

```
void addNewcrumb(
  String uid,
  String goalTitle,
  String description,
  String rule,
  String way,
  String startDate,
  String endDate,
  String reminde,
  ) async {
    // A goal entry.
    final postData = {
        'title':goalTitle
        'description': goalTitle
        'rule':rule
        'way':way
        'startDate': startDate
        'endDate': EndDate
        'remind': remind
        'crumbTitle': crumbTitle
      }

};
```

- we use "Update function" to Update data for goal node and allow user get new records from firebase database After Updating and get attributes with new values such as (title , way and reminder ...) And a message will say that "New records Updated successfully"

```
updateGoale(String ID,String goalID){
   Firestore.instance.collection('goals').document(ID).setData({
     'title':GoalCubit.get(context).goalTitleController.text
     'description': GoalCubit.get(context).goalTitleController.text
     'rule':GoalCubit.get(context).rule
     'way': GoalCubit.get(context).gwayController.text,
     'startDate': GoalCubit.get(context).goalStartDateController.text
     'endDate': GoalCubit.get(context).goalEndDateController.text
     'remind': GoalCubit.get(context).remindSelectedValue
     'crumbTitle': GoalCubit.get(context).remindSelectedValue
     ''
   }).then((value){
     print('record updated successflly');
   });
 }
```

- we use "Update function" to Update data for goal node and allow user get new records from firebase database After Updating and get attributes with new values such as (title , way and reminder ...) ,and a message will say that "New records Updated successfully"

```
updateCrumb(String goalID,String crumbTitle){
    Firestore.instance.collection('crumbs').document(ID).setData({
      'title':GoalCubit.get(context).goalTitleController.text
      'description': GoalCubit.get(context).goalTitleController.text
      'rule':GoalCubit.get(context).rule
      'way': GoalCubit.get(context).gwayController.text,
      'startDate': GoalCubit.get(context).goalStartDateController.text
      'endDate': GoalCubit.get(context).goalEndDateController.text
      ''
    }).then((value){
      print('record updated successflly');
    });
  }
```

Chapter 5: Testing

In this chapter we will test the system. All modules/components are integrated To check whether the system is working as expected or not, this plays an important role in Provide a high quality product.

5.1 Unit Testing

- Testing of individual items (e.g modules , programs , objects and classes) usually as part of the coding phase .
  in isolation from other development items and the system as a whole .

5.2 Integrated Testing

- Testing The interfaces between major (systems level application modules) and minor (individual programs or components) items within an application which must interact with each other .

5.3 Additional Testing

- Test cases :
  a- login test case
  Test scenario Objective :
  Verify log in successfully with correct email and correct password
  Assumptions – Dependencies :
  valid email : Rawan@gmail.com
  valid password : Rw12345

ID	Description	Expected	Actual	State
1	User signs up with invalid email format	An Error message saying (Enter a valid Email)	as expected	pass
2	User sign up with email which is already exist	An error message saying " This email is already exist "	as expected	pass
3	User signs up with names informative	An error message saying " This email is already exist "	as expected	pass
4	User signs up with blank name	An error message saying " please Enter your name "	as expected	pass
5	User signs up with blank password	An error message saying "please Enter your password"	as expected	pass
6	User signs up with valid data	A message saying " sign up is succeeded " and send verification link through email	as expected	pass
7	User signs up with password not in this format of 8 characters	An error message saying " please Enter your password in a format of 8 characters at minimum "	as expected	pass
8	User log in with invalid email format	An error message saying " please enter a valid email "	as expected	pass
9	User log in with valid email and incorrect password	An error message saying " please enter a correct password "	as expected	pass
10	User log in with blank email	An error message saying " please enter an email "	as expected	pass
11	User log in with blank password	An error message saying " please enter a password "	as expected	pass

12	User log in without making his account verification code	An error message " please complete the sign up process "	as expected	pass
13	User log in with valid data	The user log in Application successfully	as expected	pass
14	User access his Account by sign process and then verify his account and log in	The registration and log in process complete successfully	as expected	pass
15	User forget his password	The password changed successfully	as expected	pass
16	User forget his password and making new password without	The view of related UI of changing password shouldn’t present until user check Email and a message will append saying him check your email	as expected	pass
17	User forget his password and making password with invalid format which is 8 characters	An error message saying " please enter a valid password "	as expected	pass
18	User change password but with incorrect old password	An error message saying " please enter a correct password "	as expected	pass
19	User change password with correct old password	The password will change successfully	as expected	pass

20	User change password and user complete confirm password which not equal new password	An error message saying " Two passwords don't match others "	as expected	pass
1	User choose node goal	The system will display the screen of goal node and in the right button (+) add new goal	as expected	pass
2	User add new goal	The system will allow the user to divide The main goal to crumbs	as expected	pass
3	User uses The crumbs goal	The system will allow the user options way to system of crumbs such as : Period way and steps way	as expected	pass
4	When the user uses a path of period way	The system will allow the user deadline for finishing The crumb goal	as expected	pass
5	When the user uses a path of steps way	The system will allow the user to follow the steps of the crumbs goals and make sure that each step has been finished or not	as expected	pass
6	User makes sure main goal finish	The system will allow two options such as : in progress and completed	as expected	pass
7	User noticed that deadline for main goal less than total time of total crumbs	A message saying " please reset the time table of deadline for main goal "	as expected	pass

8	When user can join to challenge node	The user get the challenge start day and the number of challengers and the deadline and the challenge rules . System update the challenge reminders to the calendar System update the challenge deadline to the calendar	as expected	pass
9	User can create his own challenge	user set the challenge name - user set the challenge rules – user set the challenge start date – user set the challenge deadline The system will allow two options to challenge such as : private or public System update the challenge reminders to the calendar System update the challenge deadline to the calendar	as expected	pass
10	If the user miss the deadline of the challenge	System send notification reminder and a message will say that " you are slothy because you missed the challenge "	as expected	pass
11	When user can join to Daily Tasks node and add new task	-User will determine if the task for today or for another day -User select the certain day -User set the task name & the task description	as expected	pass

		-User determines if enable end date for the task or not 0 User add task reminder System update the task day to the database System update the task reminder to the calendar Task add in the list .		
12	when system allow to user the gratitude time node	-System send notification on day and night time to remind the user -User can add sentences to the gratitude time list	as expected	pass
13	User miss the gratitude time	system let the slots empty as red color and give the user notification to complete it	as expected	pass
14	user doesn't check the checklist of the task	-System update the task to unfinished task -System send notification to the user to reset the task deadline	as expected	pass
15	User can join to Routine node	User set routine title – user add tasks for routine – user set reminders – user set task time the system remind user for tasks and user check that tasks finished or not in check box – system determines the user monthly productivity	as expected	pass

16	User miss some of the tasks and finished some	- system remind The user for missed tasks - system ask user to check the missed tasks - system calculate the productivity percentage	as expected	pass
17	when system allow to user the gratitude time node	-User can add new note -User set the note title -User set a color for the note -User can add a note a group	as expected	pass
18	User add an already exist title	System will send an error message saying " write different title "	as expected	pass

Chapter 7: Results and Dicussion

In this chapter we are going to find out the results of the project whether they are achieved or not and the differences between the desired results and the actual ones.

6.1.1 Expected result

- User can register and login successfully.
- The user can add new task successfully
- The user can reach the events on the calendar successfully
- User can add new goal successfully
- User can add new Routine successfully
- User can add new note successfully
- User receive reminders successfully
- User apply the rules successfully
- User can add sentences to the gratitude list successfully
- User can see the monthly and weekly productivity progress insights

6.1.2 Actual results

- The app register and login successfully.
- The app can add new task successfully
- The app can reach the events on the calendar successfully
- The app add new goal successfully
- The app add new Routine successfully
- The app add new note successfully
- The app sent reminders successfully
- The app apply the rules successfully
- The app add sentences to the gratitude list successfully

6.2 Discussion

- We manged to get the same expected results except that the insights section

- We managed to fix that with flutter analytics package to give the user responsive data visualization

Chapter 7: Conclusion

In this chapter, we will summarize the idea of the application as a whole in the simplest lines and summarize the points mentioned above.

- **First paragraph**

  - While there are many apps for habit tracking and time management , our app has some advantages over them .

  - Through our app user can easily manage their tasks and add events at the calendar.

  - No app subscription
  - No limitations on the user data and storage

  - We use simple design and district system to easily make the user able to reach their targets

- **Second paragraph**

  - Enhanced User Experience (UX): I would invest in refining the app's user interface (UI) and navigation to provide a seamless and intuitive experience. This includes simplifying the onboarding process, improving task creation and management, and ensuring smooth interactions throughout the app.

  - Accessibility Considerations: the app will be accessible to a wide range of users, including those with disabilities. This would involve following accessibility guidelines, such as providing support for screen readers, adjustable font sizes, and color contrast options.

  - Intelligent Task Suggestions: I would leverage machine learning and data analysis to offer intelligent task suggestions to users. The app could analyze users' task history, patterns, and priorities to provide personalized recommendations for task management and help users stay organized.

Chapter 8: Future work

In this chapter, we will mention the most important points that will be developed in the future, and we will suggest what future plans we have regarding our application, as it will not end with the end of the discussion.

- Since the project’s aim is to solve the confusion problem of how to be more productive? We will keep on developing it and continue in improving it's performance and service , We will provide some features Such as:

    - Link Organizer : user can share their url links for social media posts and videos and the application automatically filters them and categorize them into folders.

    - Ai assistance: an chat bot and ai assistance to help the user reach their goals and targets on time

    - Community: To make the users able to share their progress and achievements and improve their Routine

Resources

1. https://docs.flutter.dev/
2. https://www.businessofapps.com/insights/how-to-design-a-habit-tracking-app-right/
3. https://flutterawesome.com/a-daily-task-manager-application-project-created-in-flutter/
4. https://pub.dev/
5. https://developer.android.com/studio/intro/?gclid=CjwKCAjwquWVBhBrEiwAt1Km
6. https://towardsdatascience.com/recommendation-systems-a-review-d4592b6caf4b

Books

1. Beginning App Development with Flutter- by Rap Payne.
2. Google Flutter Mobile Development Quick Start Guide- by Prajyot Mainkar and Salvatore Girodano.
3. Pragmatic Flutter- by Priyanka Tyagi.
4. Programming Flutter- by Carmine Zaccagnino.